SOFTWARE REQUIREMENTS SPECIFICATION

Fit for Surgery – Digital Prehabilitation Platform (MVP)

Document Item	Details
Purpose	Development reference for Phase 1 / MVP implementation
Scope	Phase 1 / MVP only
Status	For Development & Client Review

1. Product Overview

A secure, cloud-based digital health platform for patients awaiting major oncological surgery. Patients participate in structured prehabilitation programmes under remote clinical supervision. The solution includes a mobile PWA for Patients and Doctors and a web-based Administration Portal.

2. User Roles

Role	Primary Responsibilities
Patient	View assigned programme/tasks, record health data, complete exercises, view progress, notifications and reports.
Doctor	Register patients, review patient information, vitals, progress and compliance, and assign/modify exercise plans.
Admin	Register/manage doctors and patients, manage exercise library/videos and access administration analytics.

3. Functional Requirements

User Authentication & User Management

- Patient Registration by Doctor
- Doctor Registration by Admin
- Admin Registration
- Secure Login
- Forgot Password
- Email Verification
- Role-Based Access Control
- Basic Profile Management
- Medical Profile Management

Patient Dashboard

- Daily Health Summary
- Assigned Daily Tasks
- Exercise Schedule
- Progress Overview
- Health Status Summary
- Recent Notifications

Exercise & Prehabilitation

- Personalized Exercise Plans
- Exercise Categories
- Walking Programmes
- Strength Exercises
- Breathing Exercises

- Stretching Exercises
- Exercise Demonstration Videos
- Exercise Completion Tracking
- Daily Exercise Checklist
- Exercise History
- Exercise Compliance Report

Daily Health Monitoring

- Blood Pressure
- Pulse Rate
- Oxygen Saturation (SpO□)
- Temperature
- Weight
- Blood Sugar (Optional)
- Pain Score
- Walking Duration
- Step Count (Manual)
- Sleep Duration
- Nutrition Intake
- Water Intake
- Mood Assessment
- Maintain historical records and display trend charts for each health parameter.

Doctor Portal

- View Patient List
- Patient Timeline
- Review Daily Vitals
- Exercise Compliance Monitoring
- Progress Monitoring
- Assign or Modify Exercise Plans
- Review Patient Alerts – Basic

Clinical Analytics Dashboard

- Patient Vital Trend Charts
- Exercise Compliance Analytics
- Daily Activity Summary
- Health Improvement Indicators
- Compliance Percentage
- Weekly Progress Summary
- Clinical Overview Dashboard

Patient Timeline

- Chronological Daily Vitals
- Chronological Exercise Completion
- Chronological Notifications

Notifications & Reminders

- Exercise Reminders
- Missed Exercise Alerts
- Inactivity Alerts
- In-App Notifications

Administration Portal

- Doctor Management including registration
- Patient Management including registration
- Exercise Library Management

- Video Management – videos uploaded to YouTube as Private and accessed through the application
- Dashboard Analytics

Consent & Privacy Management

- Digital Consent Forms
- Privacy Policy Acceptance
- Research Participation Consent

Reports & Progress Summary

- Weekly Progress Reports
- Monthly Progress Reports
- Exercise Compliance Reports
- Vital Trend Reports

4. Critical Role-Based Workflows

Workflow	Expected Flow
Doctor → Patient	Doctor logs in → registers/selects patient → patient account/profile is available → patient logs in.
Doctor → Exercise	Doctor selects patient → assigns/modifies exercise plan → patient logs in → patient sees the current plan assigned to that patient.
Patient Exercise Access	Patient can access exercises assigned to the patient; unassigned exercise-library content must not be exposed as the patient's assigned exercises.
Exercise Completion	Patient opens assigned exercise → completes exercise → completion is recorded → Doctor can review compliance.
Health Monitoring	Patient enters supported health data → saves → data is retained in history → trends are available → Doctor can review daily vitals.
Patient Timeline	Vitals/exercise completion/notifications occur → supported events are retained → timeline displays them chronologically.
Role Access	Patient, Doctor and Admin access only functions permitted for their role; unauthorized functions are blocked.
Data Isolation	A patient must not see another patient's health, exercise, timeline, report or notification information.

5. Key Business Rules

- Patients view and interact with exercise plans assigned to them by a Doctor.
- Patients must not access another patient's data.
- Role-based access restricts Patient, Doctor and Admin functions.
- Exercise programmes are provided by the clinical team.
- Patients manually enter health measurements in Phase 1 unless future device integration is introduced.
- Historical health records and trend charts are maintained for supported health parameters.
- Exercise completion is associated with the correct patient and assigned exercise.
- Private YouTube video content for exercise demonstration is accessed through the application.

6. Technical Requirements

Area	Requirement
Mobile	Flutter PWA for Android
Web Portal	Next.js / React.js
Backend	NestJS (Node.js)
Database	PostgreSQL / MySQL as specified in source scope
Authentication	Firebase Authentication / Auth0
Push Notifications	Firebase Cloud Messaging (FCM)
Analytics	Chart.js
Hosting	AWS Cloud
Connectivity	Internet connectivity required for synchronization
Security	Secure cloud platform, authenticated protected functions and role-based access

7. Client Dependencies / Inputs

- AWS Account
- Firebase Project
- Domain Name and DNS Access
- SSL Certificate
- Cloud Storage Access
- Production Server
- Firebase Authentication
- Firebase Cloud Messaging (FCM)
- SMTP / Email Service
- Branding Assets
- Exercise Videos / Links
- Hospital Administrator Details
- Doctor Details, if available
- Patient Registration Form
- Alert Thresholds
- User Roles & Permissions
- Consent Forms
- Privacy Policy
- Terms & Conditions

8. Phase 1 Deliverables

- PWA Mobile Application (Android & iOS)
- Web-based Portal
- Backend APIs
- Database
- Cloud Deployment
- Database Schema

9. Explicitly Out of Scope

- High Risk Patient Identification – Phase 2
- Surgery Readiness Score (Rule-Based) – Phase 2
- HIS integration
- EMR/EHR integration
- Medical-device integrations introduced in future phases
- Video calling and chat; third-party communication services require separate licensing where applicable
- Any feature not listed in this Phase 1 SRS

The source scope explicitly marks High Risk Patient Identification and Surgery Readiness Score as Phase 2, and HIS/EMR/EHR integrations as outside the Phase 1 MVP scope.

10. Assumptions

- Patients have Android smartphones with internet connectivity.
- Exercise programmes and clinical content are provided by the clinical team/client.
- Patients manually enter health measurements in Phase 1.
- Internet connectivity is required for synchronization.
- Client provides branding and clinical content before implementation.
- Required third-party accounts/access are provided by the client.

11. Scope Change

Any feature, workflow, integration, report, design requirement or technical requirement not explicitly included in this SRS is outside the approved Phase 1 scope and should be handled through a separate change request with effort, timeline and/or commercial impact.

12. Testing & Acceptance

Development testing and UAT will be performed against the approved Phase 1 requirements. Functionality will be evaluated against this SRS and approved workflows. Requests for functionality not defined in this document are scope changes rather than defects.

14. Developer Notes

- Do not implement Phase 2 decision/analytics features as Phase 1 requirements.
- Do not expose the full exercise library to patients as a substitute for doctor assignment.
- Maintain strict patient-level data isolation.
- Enforce role-based access at UI and backend/API authorization levels.
- Keep health history and exercise completion linked to the correct patient.
- Do not assume HIS/EMR/EHR integration in Phase 1.
- Where the source scope does not define an exact validation rule, threshold, field or calculation formula, obtain client/clinical confirmation before implementation.